

Nonlinear Regression: Change of Slope and Sinusoidal Models

STAT 429: Introduction to Time Series

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1 The Change of Slope Model

We shall next start our discussion on models in which certain parameters appear in a non-linear fashion. A simple example is:

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t - c) + \epsilon_t \quad (1)$$

with $\epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$. Here $\text{ReLU}(t - c) = (t - c)_+$ equals 0 if $t \leq c$ and equals $t - c$ if $t \geq c$. We can also write

$$\text{ReLU}(t - c) = (t - c)_+ = (t - c)I\{t > c\} = \max(t - c, 0).$$

$(\cdot)_+$ is also called the positive part function, or, the ramp function.

The model (1) says that for times $t \leq c$, the slope of the regression line is β_1 , while for $t > c$, the slope changes to $(\beta_1 + \beta_2)$. We shall refer to (1) as the 'Change of Slope' model. An alternative name for this model is "Broken-stick regression". This is because the function

$$t \mapsto \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t - c)$$

resembles a broken stick.

The unknown parameters for this model are $c, \beta_0, \beta_1, \beta_2$ as well as σ . The unknown parameter c makes (1) a nonlinear regression model. If c were known, then (1) would be a linear regression model:

$$y = X_c \beta + \epsilon \quad (2)$$

with

$$X_c = \begin{pmatrix} 1 & 1 & \text{ReLU}(1 - c) \\ 1 & 2 & \text{ReLU}(2 - c) \\ 1 & 3 & \text{ReLU}(3 - c) \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ 1 & n & \text{ReLU}(n - c) \end{pmatrix} \quad \text{and } \beta := \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} \quad \text{and } \epsilon = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \cdot \\ \cdot \\ \epsilon_n \end{pmatrix}$$

1.1 Parameter Estimation

Least squares again is the most basic estimation procedure. The sum of squares is:

$$S(\beta_0, \beta_1, \beta_2, c) := \sum_{t=1}^n (y_t - \beta_0 - \beta_1 t - \beta_2 \text{ReLU}(t - c))^2. \quad (3)$$

We need to minimize this over all the four variables $\beta_0, \beta_1, \beta_2, c$. Using matrix notation, we can write

$$S(\beta, c) = \|y - X_c \beta\|^2.$$

If we fix c , then it is easy to minimize $S(\beta, c)$ over β . This is the same as linear regression and the minimizing β is given by:

$$\hat{\beta}(c) := (X_c^T X_c)^{-1} X_c^T y, \quad (4)$$

and the smallest value of $S(\beta, c)$ for fixed c is $S(\hat{\beta}(c), c)$ which is just the RSS in the multiple linear regression with fixed c . We use the notation:

$$RSS(c) = S(\hat{\beta}(c), c) = \min_{\beta} S(\beta, c). \quad (5)$$

The least squares estimates of β and c can therefore be found in the following way:

1. Fix a finite set of possible values of c . In this change of slope model, it is reasonable to assume that $c \in \{1, \dots, n\}$. One can also take a finer grid of values in the range $[1, n]$.
2. For each value of c in the chosen set, calculate $\hat{\beta}(c)$ and define $RSS(c) = S(\hat{\beta}(c), c)$.
3. Take $\hat{c}$ to be the value of c which minimizes $RSS(c)$.
4. Take $\beta = \hat{\beta}(\hat{c})$.

1.2 Bayesian Inference

For uncertainty quantification, let us consider Bayesian inference for this model. We need to first write down the likelihood and prior, and then use them to derive the posterior. For the likelihood, we shall assume (just as in usual linear regression) that the normal distribution for the errors $\{\epsilon_t\}$ in (1): $\epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$. This leads to the likelihood:

$$\prod_{t=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_t - \beta_0 - \beta_1 t - \beta_2(t-c)_+)^2}{2\sigma^2}\right) = (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) \quad (6)$$

where again $S(\beta, c)$ is the sum of squares (3).

For the prior, recall that, for linear regression, we used:

$$\beta_0, \beta_1, \beta_2, \log \sigma \stackrel{\text{i.i.d.}}{\sim} \text{unif}(-\infty, \infty).$$

Here $\text{unif}(-\infty, \infty)$ can be thought of as short form for $\text{unif}(-C, C)$ for a very large constant C . We will use the same prior here also. However we also need to supply a prior for the parameter c . Here note that c needs to be in the range $(1, n)$ (open interval from 1 to n). Because if $c \geq n$, then $\text{ReLU}(t - c) = 0$ for all $t = 1, \dots, n$ so the term $\text{ReLU}(t - c)$ is not needed in the model (1). If $c \leq 1$, then $\text{ReLU}(t - c) = t - c$ for all $t = 1, \dots, n$ so that the ReLU term in (1) becomes linear so the model again becomes a linear function of t .

Because $c \in (1, n)$, a natural prior is to take:

$$c \sim \text{uniform}(1, n). \quad (7)$$

In practice, we would not be interested in values of c that are near the boundaries 1 and n . This is because if c is very close to 1 or n , then the model is essentially linear for most of dataset. To reflect this, we can take the prior for c to be uniform in smaller range than $(1, n)$ ignoring points near each end; for example, uniform on $(5, n - 5)$. Let us work with the prior (7), and we shall mention the changes that need to be made to the posterior if we are instead working with the uniform prior on a smaller range such as $(5, n - 5)$.

The posterior joint density of all the parameters β, c, σ (here β denotes the vector consisting of $\beta_0, \beta_1, \beta_2$) is given by

$$\text{posterior}(\beta, c, \sigma) \propto \text{likelihood} \times \text{prior}.$$

The likelihood is given in (6) and the prior is

$$\begin{aligned} \text{prior density} &= \frac{I\{-C < \beta_0, \beta_1, \beta_2, \log \sigma < C\}}{(2C)^4 \sigma} \frac{I\{1 < c < n\}}{n-1} \\ &\propto \frac{I\{-C < \beta_0, \beta_1, \beta_2, \log \sigma < C, 1 < c < n\}}{\sigma} \end{aligned}$$

We shall drop the indicator terms involving C because C is very large (think ∞). This will lead to

$$\text{prior density} \propto \frac{I\{\sigma > 0, 1 < c < n\}}{\sigma}$$

The posterior is then given by

$$\text{posterior}(\beta, c, \sigma) \propto \sigma^{-n-1} \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) I\{\sigma > 0\} I\{1 < c < n\}.$$

To get the posterior density of c alone (c is the most important parameter in the model (1)), we need to integrate the joint posterior density above with respect to β and σ :

$$\text{posterior}(c) \propto I\{1 < c < n\} \int_0^\infty \sigma^{-n-1} \int \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) d\beta d\sigma.$$

Let us first calculate the inner integral. $S(\beta, c)$ is a quadratic function in β so that

$$\int \exp(-S(\beta, c)/(2\sigma^2)) d\beta$$

should be related to the normalizing constants in the multivariate normal density. To figure the integral precisely, we first use the Pythagorean identity (discussed previously):

$$\begin{aligned} S(\beta, c) &= S(\hat{\beta}(c), c) + \left(\beta - \hat{\beta}(c)\right)^T X_c^T X_c \left(\beta - \hat{\beta}(c)\right) \\ &= RSS(c) + \left(\beta - \hat{\beta}(c)\right)^T X_c^T X_c \left(\beta - \hat{\beta}(c)\right) \end{aligned}$$

where $\hat{\beta}_c$ is given in (4). The value $S(\hat{\beta}_c, c)$ is equal to $RSS(c)$ as noted in (5). Thus

$$\begin{aligned} \int \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) d\beta &= \int \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) \exp\left(-\frac{\left(\beta - \hat{\beta}(c)\right)^T X_c^T X_c \left(\beta - \hat{\beta}(c)\right)}{2\sigma^2}\right) d\beta \\ &= \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) \int \exp\left(-\frac{\left(\beta - \hat{\beta}(c)\right)^T X_c^T X_c \left(\beta - \hat{\beta}(c)\right)}{2\sigma^2}\right) d\beta \\ &= \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) (\sqrt{2\pi})^p \sqrt{\det(\sigma^2(X_c^T X_c)^{-1})} \\ &= \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) (\sqrt{2\pi})^p \sigma^p |X_c^T X_c|^{-1/2} \end{aligned}$$

where $|X_c^T X_c| = \det(X_c^T X_c)$. Here $p = 3$ because there are three components inside β . Therefore

$$\begin{aligned} \text{posterior}(c) &\propto I\{1 < c < n\} \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) (\sqrt{2\pi})^p \sigma^p |X_c^T X_c|^{-1/2} d\sigma \\ &\propto I\{1 < c < n\} |X_c^T X_c|^{-1/2} \int_0^\infty \sigma^{-n+p-1} \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) d\sigma. \end{aligned}$$

The change of variable $\sigma = s\sqrt{RSS(c)}$, gives

$$\begin{aligned} \text{posterior}(c) &\propto I\{1 < c < n\} |X_c^T X_c|^{-1/2} \left(\frac{1}{RSS(c)}\right)^{(n-p)/2} \int_0^\infty s^{-n+p-1} \exp\left(-\frac{1}{2s^2}\right) ds \\ &\propto I\{1 < c < n\} |X_c^T X_c|^{-1/2} \left(\frac{1}{RSS(c)}\right)^{(n-p)/2}. \end{aligned} \tag{8}$$

This posterior will be evaluated numerically over a grid of values of c in the range $(1, n)$. The term $|X_c^T X_c|^{-1/2}$ will become infinite when $|X_c^T X_c| = 0$ i.e., when X_c does not have full column rank. This is the case when c is outside of the range $(1, n)$. When c is in the range $(1, n)$, the determinant of $X_c^T X_c$ will be non-zero but it will still be good to not consider c too close to 1 or n to prevent numerical instability due to near-singularity.

If the prior is taken to be uniform on a different range (say $(5, n - 5)$), then we simply need to change the indicator $I\{1 < c < n\}$ in (8) by the indicator over the prior range.

The main term in the posterior (8) is $(1/RSS(c))^{(n-p)/2}$ (the other term $|X_c^T X_c|^{-1/2}$ generally does not vary significantly with c). This term takes its largest value when c equals the least squares estimator $\hat{c}$ (note $\hat{c}$ minimizes $RSS(c)$). The size of the power $n - p$ determines the amount of concentration of the posterior around $\hat{c}$. When $n - p$ is large, the posterior is very tightly concentrated around $\hat{c}$.

2 The Sinusoidal Model

By the sinusoidal model, we refer to:

$$y_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) + \epsilon_t \quad \text{where } \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2). \quad (9)$$

This is a simple model for time series data which show strong periodicity (like the sunspots data).

The unknown parameters in this model are $\beta_0, \beta_1, \beta_2, \sigma$ as well as the frequency parameter f . If f is assumed to be known, then clearly (9) is a multiple linear regression model:

$$y = X_f \beta + \epsilon$$

with

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \quad \text{and} \quad X_f = \begin{pmatrix} 1 & \cos(2\pi f(1)) & \sin(2\pi f(1)) \\ \vdots & \vdots & \vdots \\ 1 & \cos(2\pi f(n)) & \sin(2\pi f(n)) \end{pmatrix} \quad \text{and} \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} \quad \text{and} \quad \epsilon = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

When f is unknown, this is a nonlinear regression model.

The function $t \mapsto \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t)$ is called a sinusoid. Before proceeding further, let us look at some basic properties and terminology related to sinusoids.

2.1 The Sinusoid

When we say sinusoid, we refer to the following function of time t :

$$s(t) := \beta_0 + R \cos(2\pi f t + \phi) \quad (10)$$

Here

- R is called the *amplitude*. It represents the height of the oscillation from its center line.
- f is called the *frequency*. It represents the number of oscillations in unit time. If time is measured in seconds, then the unit of f is Hertz (Hz).

- $1/f$ is called the *period*. It is the length of time to complete one full oscillation.
- ϕ is called the *phase*. Without ϕ (i.e., if $\phi = 0$), then the above sinusoid becomes $\beta_0 + R \cos(2\pi ft)$ so it starts at its maximum value at $t = 0$. Adding ϕ shifts the wave left or right in time. This captures the fact that different oscillations might 'start' at different points in their cycle.
- $2\pi f$ is called the *angular frequency*. Sometimes we use the notation $\omega = 2\pi f$ for the angular frequency. It measures the rate of change of the angle inside the cosine.

Using the formula $\cos(\alpha + \beta) = (\cos \alpha)(\cos \beta) - (\sin \alpha)(\sin \beta)$, we can represent the sinusoid (10) in the following equivalent alternative form:

$$s(t) = \beta_0 + \beta_1 \cos 2\pi ft + \beta_2 \sin 2\pi ft. \quad (11)$$

The parameters β_1, β_2 in (11) are related to R, ϕ in (10) via $\beta_1 = R \cos \phi$ and $\beta_2 = -R \sin \phi$. While working with models involving sinusoids, we use the representation (11) because the parameters β_1 and β_2 appear linearly in (11).

2.2 Discrete sampling and restricting f to $[0, 1/2]$

Often in time series analysis, we work with equally spaced time points and assume that the time variable t takes the values $1, \dots, n$ (where n is the sample size). It turns out that if we consider the sinusoid (10) and restrict the time t to $1, \dots, n$, then we can always constrain the frequency parameter f to $[0, 1/2]$. This is a consequence of the following result.

Fact 2.1. *For every $f \in (-\infty, \infty)$ and $\phi \in (-\infty, \infty)$, there exists $f_0 \in [0, 1/2]$ and $\phi_0 \in (-\infty, \infty)$ such that*

$$s(t) = \beta_0 + R \cos(2\pi ft + \phi) = \beta_0 + R \cos(2\pi f_0 t + \phi_0) \quad \text{for all } t = 1, \dots, n.$$

Proof. Consider the following three cases.

1. If $f < 0$, then we can write $\cos(2\pi ft + \phi) = \cos(2\pi(-f)t - \phi)$. Clearly, $-f \geq 0$.
2. If $f \geq 1$, then we write (below $[f]$ is the largest integer less than or equal to f):

$$\cos(2\pi ft + \phi) = \cos(2\pi[f]t + 2\pi(f - [f])t + \phi) = \cos(2\pi(f - [f])t + \phi),$$

because $\cos(\cdot)$ is periodic with period 2π . Clearly $0 \leq f - [f] < 1$.

3. If $f \in [1/2, 1)$, then

$$\cos(2\pi ft + \phi) = \cos(2\pi t - 2\pi(1 - f)t + \phi) = \cos(2\pi(1 - f)t - \phi)$$

because $\cos(2\pi t - x) = \cos x$ for all integers t . Clearly $0 < 1 - f \leq 1/2$.

Thus the sinusoid $R \cos(2\pi ft + \phi)$ equals $R \cos(2\pi f_0 t + \phi_0)$ at all integers t for some $0 \leq f_0 \leq 1/2$ and a phase ϕ_0 that is possibly different from ϕ . $\square$

From now on, when we discuss sinusoids $s(t) = \beta_0 + R \cos(2\pi ft + \phi)$ in the context of $t = 1, \dots, n$, we shall assume that the frequency parameter f is restricted to $[0, 1/2]$. Note also the behavior of the sinusoid for the two frequency extremes $f = 0$ and $f = 1/2$. When

$f = 0$, the sinusoid $s(t)$ is simply a constant function equal to $\beta_0 + R \cos(\phi)$. When $f = 1/2$, we have

$$s(t) = \beta_0 + R \cos(\pi t + \phi) = \beta_0 + R(\cos \phi) \cos(\pi t) = \beta_0 + R(-1)^t \cos \phi.$$

This sinusoid exhibits the maximum possible oscillation going back and forth between $\beta_0 + R \cos \phi$ and $\beta_0 - R \cos \phi$.

2.3 Least Squares Estimation of β, f, σ

We use exactly the same method for parameter estimation as in the change of slope model. We take a bunch of possible values of f , calculate the goodness of fit $RSS(f)$ of the resulting linear regression model with fixed f , and then use $\hat{f}$ as the minimizer of $RSS(f)$ over f . The remaining parameters (β and σ) are estimated as in usual linear regression with f fixed at $\hat{f}$. This algorithm is described below:

1. Take a grid of all possible values of f in the range $[0, 1/2]$.
2. For each frequency value f in the grid,
 - a) Form the matrix X_f
 - b) Do a regression of y on X_f and compute the Residual Sum of Squares $RSS(f)$
3. Take $\hat{f}$ to be the grid value which minimizes $RSS(f)$ over all the grid values.
4. Take $\hat{\beta}$ and $\hat{\sigma}$ to be the usual regression estimates (of β and σ) in the linear regression of y on $X_{\hat{f}}$.

2.4 Bayesian Posterior

For Bayesian inference, we need to select the prior on β, σ and f . For β and σ , we shall use, as usual:

$$\beta_0, \beta_1, \beta_2, \log \sigma \stackrel{\text{i.i.d}}{\sim} \text{unif}(-C, C).$$

For f , we shall use:

$$f \sim \text{unif}[0, 1/2]$$

because, as seen in Section 2.2, we know we can restrict f to $[0, 1/2]$.

This will let us write the joint posterior of all the parameters β, σ, f , and then we integrate out β and σ to deduce the posterior of f . This calculation is exactly the same as for the change of slope model; the only difference being that c there is now replaced by f (also the indicator $I\{1 < c < n\}$ should be replaced by $I\{0 \leq f \leq 1/2\}$). The posterior for f will be given by:

$$\text{posterior}(f) \propto I\{0 \leq f \leq 1/2\} |X_f^T X_f|^{-1/2} \left(\frac{1}{RSS(f)} \right)^{(n-p)/2}.$$

This posterior will be evaluated numerically over a grid of values of f in the range $[0, 0.5]$. The term $|X_f^T X_f|^{-1/2}$ becomes infinite when $|X_f^T X_f| = 0$ i.e., when X_f does not have full column rank. This will be the case when $f = 0$ or $f = 1/2$. We will exclude these edge cases while computing this posterior:

$$\text{posterior}(f) \propto I\{0 < f < 1/2\} |X_f^T X_f|^{-1/2} \left(\frac{1}{RSS(f)} \right)^{(n-p)/2}. \quad (12)$$